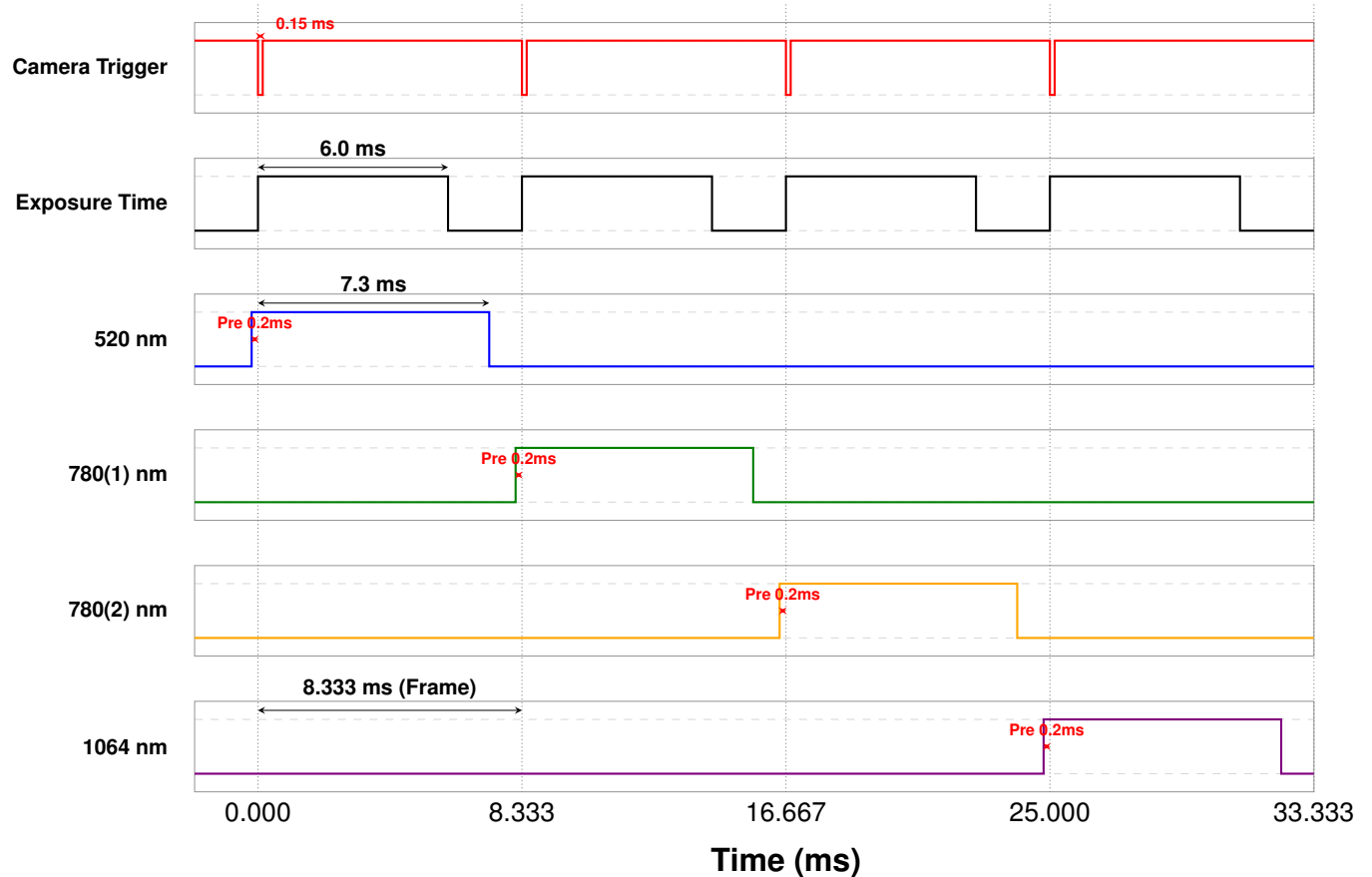


Arduino Timing Diagram (6.0 ms Exposure)

Timing Diagram Overview

Here is the logic graph for the 120 Hz firmware with Pre-trigger. The camera exposure time is set to **6.0 ms**, while the total LED ON time is **7.5 ms**.

Timing Diagram - Pre 0.2ms, Exposure 6.0ms, LED ON 7.5ms



Signal Details & Timings

- **Frame Rate:** 120 Hz (1 frame = ~8.333 ms).
- **Camera Trigger** ($t = 0.00$ ms): The trigger signal pulls LOW for **0.15 ms** (150 μ s) to command the camera to expose.

- **Control (LED):** Turns ON at $t = -0.20$ ms (**0.2 ms before** the trigger fires) to allow light stabilization, and stays ON until $t = 7.30$ ms.
- **Camera Exposure:** Starts exactly when the trigger fires ($t = 0.00$ ms), and lasts for **6.0 ms**, ending at $t = 6.00$ ms.
- **Total ON Time:** Exactly **7.5 ms** (0.2 ms pre-trigger + 7.3 ms post-trigger), ensuring it covers the full 6.0 ms exposure period safely.